

99.500: Guidelines for the Mid-Term Project

General

1. The mid-term assessment consists of: (a) creating a mock exam, (b) providing solutions, and (c) an oral presentation. It is a take-home, open-resource, group project.
2. Students are responsible for goal-oriented learning, ensuring alignment with module learning outcomes and their own expectations.

Important Dates

3. 1st submission of “exam report” by **23:59, 13/03/2026 (Recess Week)**.
4. Final submission of all the files (“exam report”, ppt slides, codes) by **23:59, 03/04/2026**.
5. Oral presentations will be held during class on **18/03/2026 (Week 8)**.

Format

6. Work in your designated homework group.
7. Each group must create two problems covering key learning outcomes, discussed topics, and personal learning expectations.
8. Each problem should have up to maximum of 5 parts, similar to homework problems.
9. Problems must have a real-world context (real or imaginary), not purely mathematical.
10. The mock exam should take an average student approximately 3 hours to complete.
11. Submit problem statements and solutions as an “exam report” (sample provided in eDimension).
12. Submit a softcopy in MS-Word (.doc/.docx) or LaTeX file. Use Times New Roman, 12 pt font for the main body. No length limit but maintain a reasonable workload.
13. If numerical methods (MATLAB/Python) are used, submit source codes properly named for evaluation.
14. Each group presents the “exam report” by one or multiple members in ppt slides. Focus on problem significance and critical/creative thinking. Presentations are 15 minutes, followed by audience questions and instructor comments.

Recommendations

15. Problems can be close-ended or open-ended (preferred).
16. Ill-defined problems are encouraged but clearly state reasonable assumptions with justifications.
17. Problems should demonstrate mastery of module concepts and may integrate interdisciplinary knowledge.
18. Further qualitative/quantitative analysis of the problem/model/solution is encouraged.

Grading

19. All group members receive the same score (0-100 scale).
20. Grading criteria: Real-world relevance of problems; Justification of assumptions; Reasonable workload (~3 hrs to solve); Correct application of learned techniques; Clarity of report; Coverage of content and techniques; Evidence of critical and creative thinking (explicitly stated in report and/or highlighted in presentation).